

OLIST E-COMMERCE DATASET: BUSINESS REPORT

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1. Project Objectives

The project aims to accomplish 2 major primary objectives.

1. To identify operational bottlenecks and revenue growth opportunities within the Olist platform to improve customer satisfaction, optimize logistics and drive profitability.
2. To create end-to-end data pipeline, including relational database design, data cleaning and data visualization.

Target Audience: Business leadership (focusing on ROI, cost containment and strategy) and Data Engineering/Analytics teams (focusing on data pipeline).

2. Data Collection and Methodology

2.1. Data Source

Olist Brazilian E-Commerce Public Dataset, containing approximately 100,000 orders, 3,000 sellers and 33,000 products spanning 2016 to 2018.

URL: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

2.2. Data Processing (MySQL)

Raw CSV files were imported into a MySQL database. The data underwent a multi-stage cleaning process (`\_staging\_1`, `\_staging\_2`) involving whitespace trimming, null handling, string-to-datetime conversions and orphan checks.

2.3. Data Modeling

Several challenges, such as 981,148 redundant geolocation coordinates and multiple review fan-outs, were resolved using Common Table Expressions (CTEs) and window functions (ROW_NUMBER ()) to ensure analytical accuracy without data loss.

2.4. Visualization (Power BI)

Cleaned schemas were imported into Power BI and utilized to track dynamic KPIs across interactive geographical and temporal filters.

3. Key Metrics and KPIs

Key metrics across 4 categories were selected to conduct the analysis.

1. **Financial:** Total Revenue (\$13.59M)
2. **Volume:** Total Orders (99.4K), Total Customers (96K), Total Active Sellers (3K)
3. **Operational:** Late Delivery Percentage, Average Delivery Days
4. **Customer Satisfaction:** Average Review Score (1-5 scale)

4. Data Analysis and Interpretation

4.1. Regional Delivery Inefficiencies

Delivery delays are not a uniform, instead concentrated around Northern and Northeastern states. Those States exhibit the highest late delivery rates, Alagoas (24%) and Maranhao (20%), shows high correlation with States with lowest sales and revenue. Distance from the country's main logistics hubs in the Southeast highly correlates with late deliveries. Such States are to receive lowest customer review scores, leading to lowest customer satisfaction.

4.2. Supply Chain Bottleneck

Analysis of the distinct delivery legs (purchase to carrier vs. carrier to customer) proves that last mile carrier logistics are the primary driver of delays. Seller dispatch efficiency is high and is not the root cause of late deliveries.

4.3. Volume vs. Quality Degradation

Revenue shows consistent, steep growth from Q1 2017 through Q2 2018. However, during the highest transaction volume months (Q4 2017 - Q1 2018), average customer review scores declined and delivery times increased.

4.4. Category Profitability vs. Satisfaction

'Health Beauty' (\$1.25M), 'Watches Gifts' (\$1.20M) and 'Bed Bath Table' (\$1.03M) lead the platform in total revenue. A combined rank analysis (revenue and satisfaction) revealed that high performing financial categories like 'Bed Bath Table', 'Office Furniture' and 'Watches Gifts' suffer from the lowest average customer review scores on the platform.

5. Strategic Recommendations

5.1. Targeted Logistics Intervention

Logistics overhauls for North and Northeast regions. Renegotiate carrier Service Level Agreements (SLAs) or onboard localized logistic partners for the North and Northeast regions (specifically Alagoas and Maranhao) to resolve last mile bottlenecks.

5.2. Predictive Capacity Scaling

The data indicates that Olist's operational capacity breaks under peak demand. Implement predictive volume forecasting ahead of Q4 surges to preemptively scale carrier capacity and customer support to prevent the satisfaction dips observed in late 2017.

5.3. High-ROI Quality Assurance

Initiate immediate seller audits and product quality reviews for 'Bed Bath Table' and 'Watches Gifts'. Because these categories already responsible for massive demand and revenue, minor improvements in product quality and fulfillment will yield a high increase in overall platform retention and customer lifetime value.